ECG-BASED ICU MORTALITY PREDICTION

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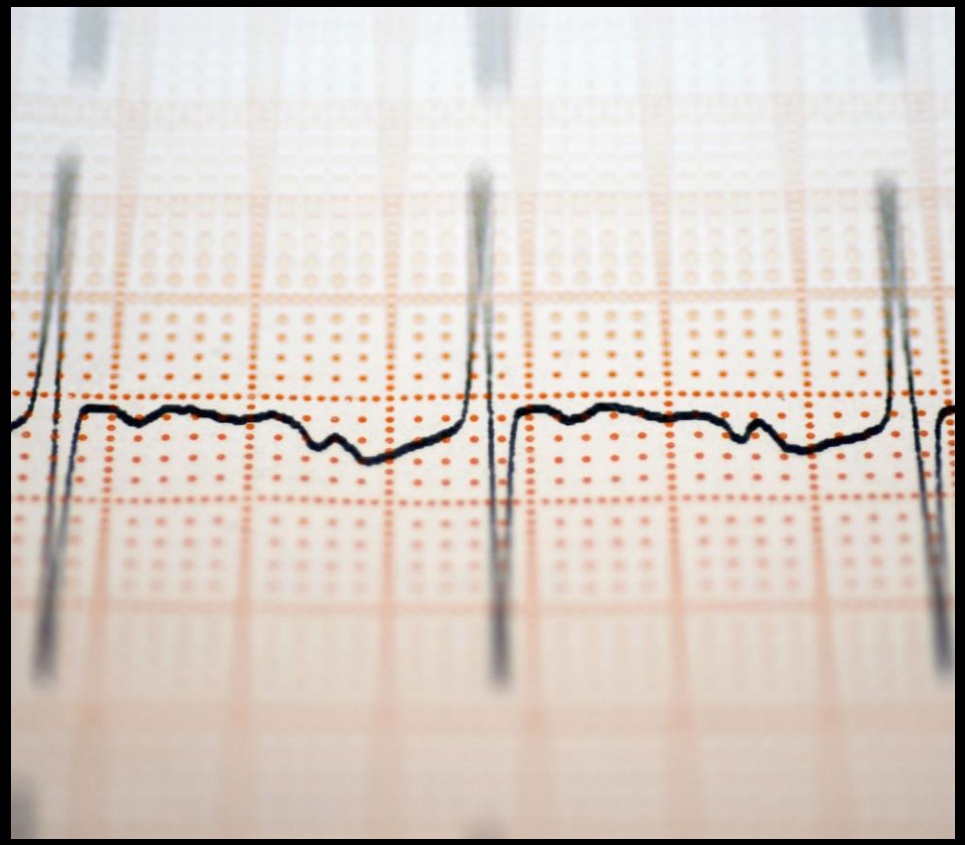
PROBLEM STATEMENT

This project investigates which ECG features are most predictive of in-hospital mortality among ICU patients using the MIMIC-IV clinical database. We build and compare multiple machine learning models (Logistic Regression, Random Forest, Gradient Boosting, and XGBoost) trained on ECG measurements, patient demographics, and vital sign extracted from BigQuery.



RESEARCH QUESTION

- Can routinely collected EHR data — including ECG rhythm, vital signs, and lab values — predict ICU mortality?
  - ICU mortality prediction currently relies on manual scoring systems like APACHE-II
  - APACHE-II requires bedside calculation of 12 variables — labor intensive and error prone
  - Cardiac rhythm data is not included in any standard severity score
  - ECGs are already collected on every ICU patient

EXISTING WORK

- Dataset source:
- MIMIC-IV v3.1: **Medical Information Mart for Intensive Care**
- **MIMIC-IV-ECG**: Diagnostic Electrocardiogram Matched Subset

CONTRIBUTION

- Jack
  - Wrote SQL queries to pull ICU admissions, ECG measurements, vital signs, and lab values from MIMIC-IV via BigQuery
  - Built ECG rhythm bucketing function using clinical domain knowledge (13 categories, priority-ordered)
  - Designed full cleaning pipeline — physiological bounds, sentinel value removal, 24h averaging, bucket-based imputation
  - Identified and removed data leakage variables (LOS, discharge time)

CONTRIBUTION

- Bob
  - Addressed severe class imbalance (~88/12) using model-specific strategies including class_weight, scale_pos_weight, and threshold tuning -> increasing death recall from near zero to 83.5%
  - Tuned XGBoost via grid search and early stopping, achieving the best overall performance (ROC AUC 0.852, PR AUC 0.501) across five trained models
  - Conducted SHAP analysis on XGBoost to identify and interpret the most predictive features of ICU mortality
  - Built reusable utility functions to improve reproducibility and team collaboration

TOOLS/METHODS USED

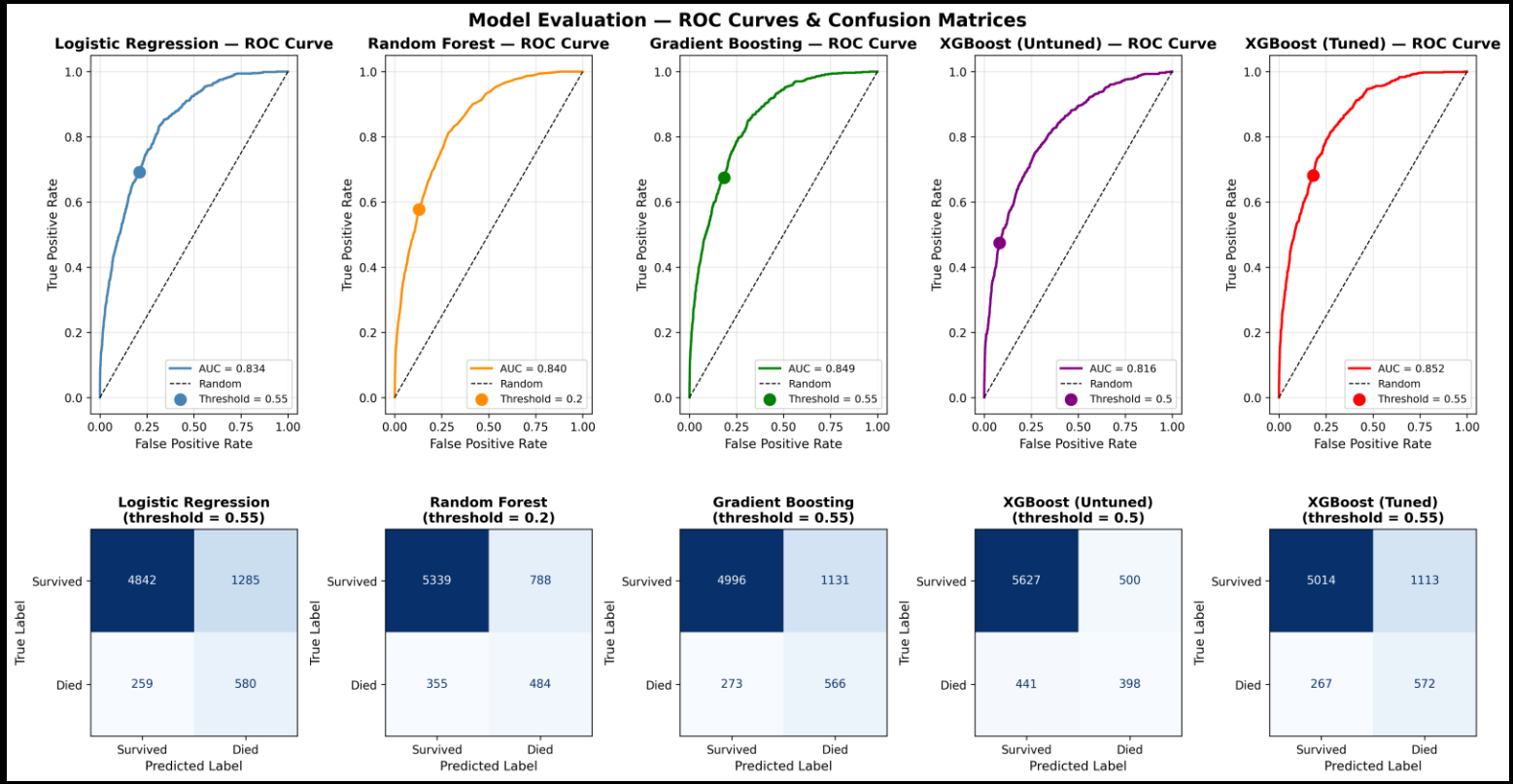
- Google BigQuery
- Python 3.13+
- Libraries:
- pandas>=1.3.0
- db-dtypes
- python-dotenv>=1.0.0
- wfdb
- regex
- matplotlib
- seaborn
- scikit-learn>=1.3.0
- numpy>=1.24.0
- shap

ISSUES OVERCOME ALONG THE WAY

- Class imbalance (88/12) meant a naive model could hit 88% accuracy by predicting "survived" for everyone — addressed using model-specific weighting (class_weight, scale_pos_weight) to prioritize deaths
- Default threshold of 0.5 still predicted no deaths even after reweighting — tuned thresholds per model (RF=0.20, XGB=0.40) to actually capture the minority class
- Used StratifiedKFold(5) to preserve the 88/12 class ratio across all cross-validation folds, preventing unreliable validation
- SHAP was too slow on the full test set — used a 500-sample subset, which is standard practice with minimal impact on interpretation

RESULTS ...

MODEL PERFORMANCE: ROC + CONFUSION MATRIX



MODEL PERFORMANCE

- All five models significantly outperform the random baseline (PR AUC ~0.12), with ROC AUC ranging from 0.816 to 0.852
- XGBoost (Tuned) achieves the best overall performance — highest ROC AUC (0.852) and PR AUC (0.501)
- Default threshold of 0.5 caused RF and GB to predict almost no deaths -> threshold tuning was essential for all models
- PR AUC is the more informative metric here -> ROC AUC can be inflated by the large number of easy-to-predict survivors

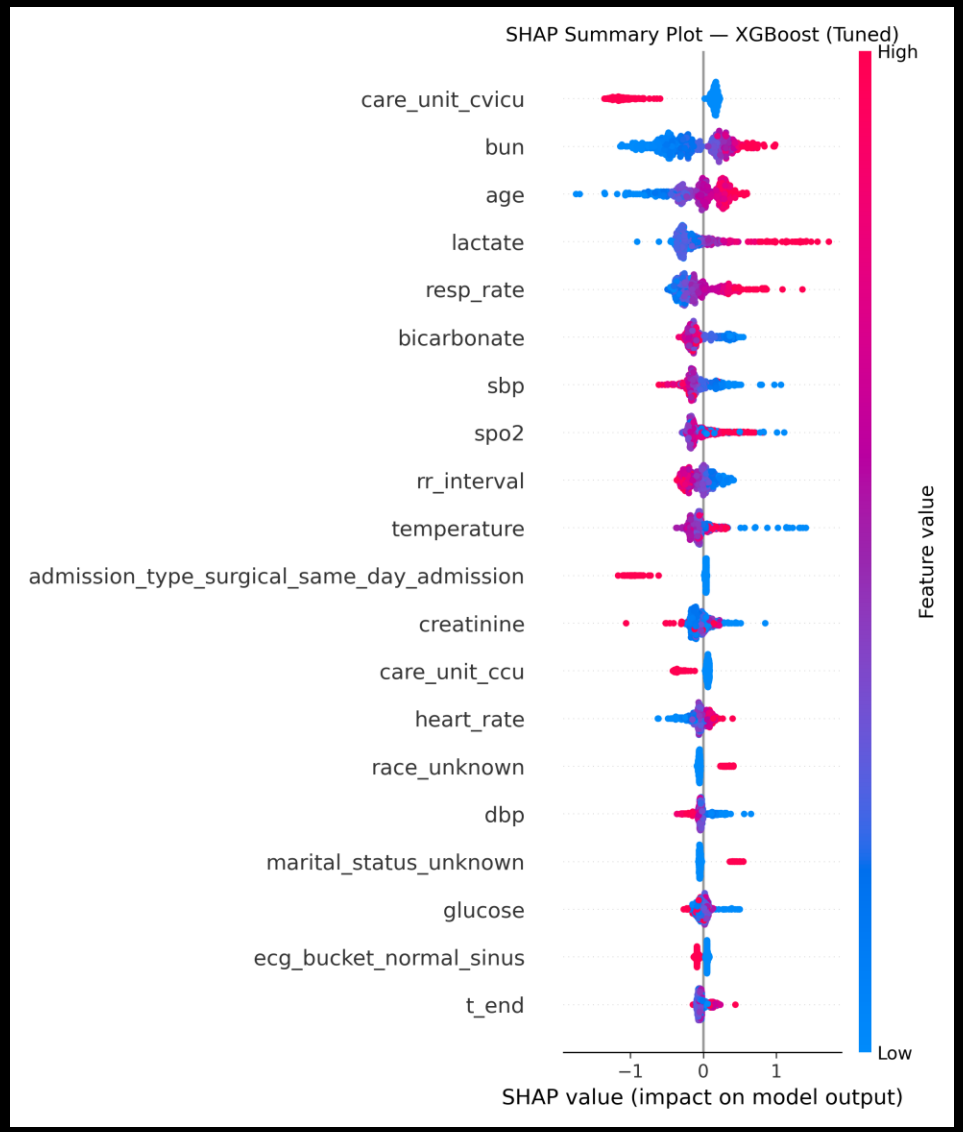
Model	ROC AUC	PR AUC	Threshold	Deaths Caught (TP)	Deaths Missed (FN)	Recall
Logistic Regression	0.834	0.458	0.55	**580/839**	259	**69.1%**
Random Forest	0.840	0.461	0.20	484/839	355	57.7%
Gradient Boosting	0.849	0.492	0.55	566/839	273	67.5%
XGBoost	0.816	0.461	0.50	398/839	441	47.4%
XGBoost (Tuned)	**0.852**	**0.501**	0.55	572/839	267	68.2%

XGBOOST (TUNED) — THRESHOLD ANALYSIS

Threshold	Deaths Caught (TP)	Deaths Missed (FN)	Recall
0.55	572/839	267	68.2%
0.4	701/839	138	83.6%

SHAP ANALYSIS

- Organ dysfunction (BUN, lactate, bicarbonate) and clinical context (CVICU, admission type) are stronger mortality predictors than any individual ECG measurement
- Among ECG features, RR interval (bradycardia) is the most predictive of in-hospital death
- Mortality risk is best captured by combining ECG measurements with vital signs and organ function markers — no single feature type is sufficient alone



CONCLUSON

- ICU mortality prediction has long relied on manual scoring systems like APACHE-II — labor intensive, error prone, and blind to cardiac rhythm data despite ECGs being collected on every ICU patient
- We showed that routinely collected EHR data alone — ECG measurements, vital signs, and lab values — can predict ICU mortality with strong discriminative performance (ROC AUC 0.852, PR AUC 0.501)
- At an optimized threshold, our best model caught 83.6% of deaths — automated, continuously updatable, and requiring no additional data collection beyond what already exists in the EHR
- SHAP analysis confirmed that combining ECG features with organ dysfunction markers outperforms any single feature type, validating the multi-modal approach
- This work demonstrates that the cardiac rhythm data clinicians have been collecting for decades contains meaningful mortality signal — it just hasn't been incorporated into risk scoring until now